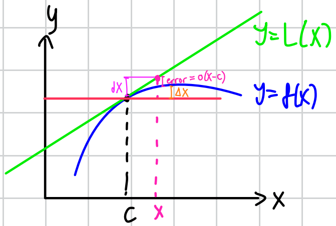
3.9 (Continue)

Suppose $f(x)$ differentiable at $c$

Define **linearization** of $f(x)$ at $x=c$ by $L(x) = f(c) + f'(c)(x-c)$




meant to approximate $f(x)$, $x \approx c$

$f(x) \approx L(x)$ for $x \approx c$

e.g. 1 $f(x) = \sin x$, $c = 0$

$\to L(x) = x$

e.g. 2 $f(x) = \tan x$, $c=0$

$\to L(x) = x \to \tan x \approx x$, $x \approx 0$

e.g. 3 $f(x) = (1+x)^k$, $c=0$, $k \in \mathbb{R}$

$$\begin{aligned} L(x) &= f(0) + f'(0)\cdot(x-0) \\ &= 1 + k(1+x)^{k-1}(1)(x) \\ &= 1 + kx\underbrace{(1+x)^{k-1}}_{\hookrightarrow (1+0)^{k-1} = 1} \\ &= 1 + kx \end{aligned}$$

Special case $k = \frac{1}{2} \to f(x) = \sqrt{1+x}$

$\sqrt{1+x} \approx 1 + \frac{x}{2}$ ; $x \approx 0$

$k = -1 \to f(x) = \frac{1}{1+x}$

$L(x) = 1 - x \approx \frac{1}{1+x}$, $x \approx 0$

$k = -\frac{1}{2} \to f(x) = \frac{1}{\sqrt{1+x}} \approx 1 - \frac{k}{2}$

e.g. 4 $\sqrt{9.01} = \sqrt{9 + 0.01} = 3\sqrt{1 + \frac{0.01}{9}} \approx 3\left(1 + \frac{0.01}{9}\right)^{1/2}$

$\approx 3\left(1 + \frac{0.01}{9}\cdot\frac{1}{2}\right)$

$$\begin{aligned} f(9.01) \approx L(9.01) &= f(9) + f'(9)(x-9)\Big|_{x=9.01} \\ &= 3 + \frac{1}{2\sqrt{9}}(9.01 - 9) \\ &= 3 + \frac{0.01}{6} \end{aligned}$$

เท่ากัน

Q: How good is the linear approximation

A: $$\frac{f(x) - L(x)}{x-c} = \frac{f(x) - f(c) - f'(c)(x-c)}{x-c}$$

$$= f'(c) - f'(c) = 0 \quad (\text{as } x \to c)$$

$$\lim_{x\to c} \frac{f(x) - L(x)}{x-c} = 0 \to \frac{f(x)-L(x)}{x-c} = o(1) \text{ as } x \to c$$

Notation: If $\lim_{x\to c} g(x) = 0$, we write $g(x) = \boxed{o(1)}$ as $x \to c$

↳ Infinitesimal: quantity converges to 0.

e.g. $\sin x = o(1)$, $x \to 0$

If $\exists c > 0$ s.t. $|g(x)| \le c$ for $x \approx c$

then write $g(x) = \boxed{O(1)}$, as $x \to c$

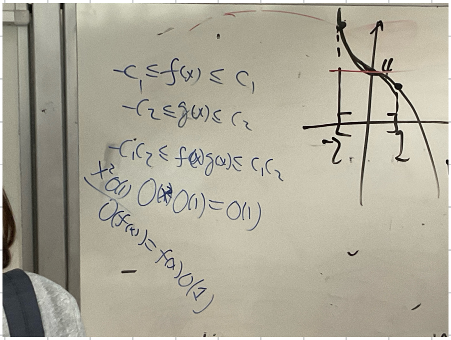
↳ Bounded

e.g.

$$o(1) + o(1) = o(1)$$
$$o(1) + O(1) = O(1)$$
$$o(1) \cdot O(1) = O(1)$$
$$O(1) + O(1) = O(1)$$
$$O(1) \cdot O(1) = O(1)$$
$$o(1) \cdot o(1) = o(1)$$
$$\frac{o(1)}{O(1)} = \text{undefined}$$

$o(1)$ $O(1)$

$(x \sin \frac{1}{x})$




More notation

$$o(h(x)) = h(x)o(1)$$
$$O(h(x)) = h(x)O(1)$$

$$f(x) = L(x) + o(1)(x-c)$$
$$= L(x) + o(x-c)$$

Recall: $\lim_{x\to 0} \frac{\sin(423x)}{\sin(821x)} = \lim_{x \to 0} \frac{423x + o(423x)}{821x + o(821x)} = \frac{423}{821}$

$o(423x) = 423 \times o(1)$, $o(821x) = 821 \times o(1)$

As $f(x) = f(c) + f'(c)(x-c) + o(x-c)$

$f(x) - f(c) = f'(c)(x-c) + o(x-c)$

$\Delta f = f'(c)\Delta x + o(\Delta x)$

write $\Delta x$ as $dx$ $\quad \approx f'(c)\Delta x$ if $\Delta x \approx 0$

Leibniz: $\frac{df}{dx} = f'$

$\Delta f \approx f'(c)\, dx$ if $dx \approx 0$

Def $f'(c)\,dx$ is called differential of $f$ at $x=c$, also written as $dy|_{x=c}$ or $df|_{x=c}$

$f'(x)dx \to$ differential of $f$ at generic $x$ — ⊛

$\to \Delta f \approx df|_{x=c}$

difference $\approx$ differential

Hence, $f(x) = L(x) + o(x-c)$

$= f(c) + f'(c)(x-c) + o(x-c)$

$= f(c) + df|_{x=c} + o(dx)$

Ex. $y = x^3 - 3\sqrt{x}$

3.9/17 $dy = (x^3 - 3\sqrt{x})'\,dx$

$= 3x^2 - 3\left(\frac{1}{2\sqrt{x}}\right)$

3.9/41 ⊛ $A = \pi r^2$

$$\Delta A \overset{\circledast}{=} dA = \frac{dA}{dr}\cdot dr = 2\pi r\,dr\Big|_{\substack{r=2\\ dr=0.02}} = 2\cdot\pi\cdot 2\cdot 0.02 \approx 0.08\pi$$

## Rigorous proof of chain rule

Suppose $g(x)$ is diff at $x=c$, $f(u)$ is diff at $u=g(c)$

$\therefore$ $f\circ g(x)$ is diff at $x=c$

$(f\circ g)'(c) = f'(g(c))g'(c)$

Since $g$ is diff at $x=c$, we have

$g(c+\Delta x) = g(c) + g'(c)\cdot\Delta x + o(x)$ as $\Delta x \to 0$

Since $f$ is diff at $u=g(c)$, we have

☺ $\to f(g(c)+\Delta u) = f(g(c)) + f'(g(c))\Delta u + o(\Delta u)$ as $\Delta x \to 0$

Let $\Delta g = g(c+\Delta x) - g(c)$, recall differentiability implies continuity

$\therefore$ $\Delta g \to 0$ as $\Delta x \to 0$

Use $\Delta u$ as $\Delta g$ in ☺ $\to$ $f(g(c+\Delta x)) = f(g(c)) + f'(g(c))\Delta g + o(\Delta g)$

$= f(g(c)) + f'(g(c))\left(g'(c)\Delta x + o(\Delta x)\right) + \Delta g\, o(1)$

$$\frac{f(g(c+\Delta x)) - f(g(c))}{\Delta x} = \frac{f'(g(c))\,g'(c)\Delta x}{\Delta x} + \frac{f'(g(c))\,o(\Delta x)}{\Delta x} + \frac{\Delta g\, o(1)}{\Delta x}$$

$$(f\circ g(x))'\Big|_{x=c} = f'(g(c))\,g'(c) + \cancel{0 + g'(c)\cdot 0}$$

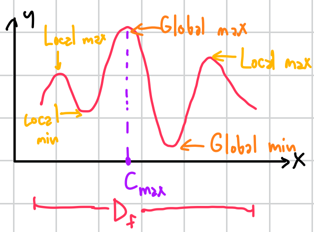
---

# Chapter 4

## 4.1 Extreme values

**Def** If $c \in D_f$ s.t. $f(c) \geq f(x)$ $\forall x \in D_f$ $\to$ $f(c)$ is the absolute/global maximum value of $f$

$f(c) \leq f(x)$ ... minimum




**Def** If $x \in D_f$ s.t. $f(c) \geq f(x)$ for $x \in D_f \cap$ open interval with $c$, we say $f(c) \to$ local max value

Q: How to find extreme values?

A: Derivative test

$\to$ Suppose $f(c)$ is local extreme value and $c \in$ interior of $D_f$ (ไม่ใช่ขอบ)

Assume $f$ diff at $x=c$, then $f'(c) = 0$

Candidates for local extremes

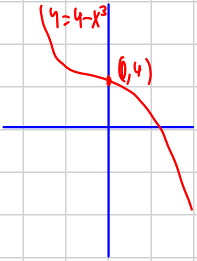
- $f'(c)=0$
- $c \in$ interior but $f'(c)$ DNE
- Boundary of $D_f$

} Critical points

Recipe:

1) Find all critical points
2) Evaluate $f$ at all CPs.
3) Evaluate $f$ at boundary pts.
4) Choose the largest $f$ value $\rightarrow$ global max
5) Choose the smallest $f$ value $\rightarrow$ global min

4.1/24 $f(x)=4-x^3, \ -2 \le x \le 1$




1. C.P.: $f'(x) \overset{\text{set}}{=} 0$

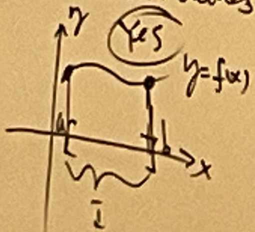
   $-3x^2=0 \rightarrow x=0$

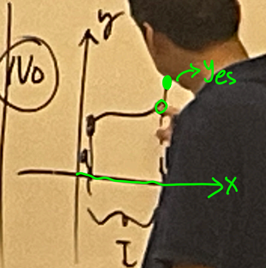
2. $f(\text{C.P.})=f(0)=4$
3. $f(-2)=\boxed{12}, \ f(1)=\boxed{3}$
4. Global max (12), Global min (3)

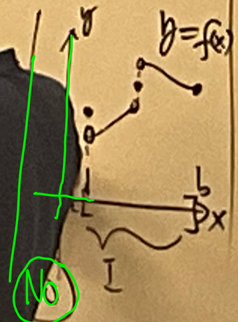
$\rightarrow$ A Critical point may not give rise to a local extreme

4.1 (Continued)

**Q**: Given a function $f(x)$ on an interval $I$, does it always have global/ ...te maximum & minimum values?




2 อันขวา ไม่ continuous

## Existence of Global Extremes

**Thm** Suppose $f(x)$ continuous on $[a,b]$, then $\exists\, x_m, x_M$ s.t.
$f(x_m) =$ global min, $f(x_M) =$ global max

(Extreme value theorem)

## 4.2 Mean Value Theorem


Suppose $f(x)$ is continuous in $[a,b]$, $f(a)=f(b)$

Assume $f$ is differentiable in $(a,b)$, then $\exists\, c \in (a,b)$ s.t. $f'(c)=0$ (Rolle's theorem)

Proof: By Extreme Value Theorem, $\exists\, x_m, x_M \in [a,b]$ s.t. $f(x_m) =$ global min value of $f$ on $[a,b]$
$f(x_M) =$ global max value of $f$ on $[a,b]$

Case 1 Either $x_m$ or $x_M \in (a,b)$

1.1 $x_m \in (a,b) \xrightarrow{\text{first derivative test}} f'(x_m)=0$ $\therefore$ Take $c = x_m$

1.2 $x_M \in (a,b) \xrightarrow{\text{first derivative test}} f'(x_M)=0$ $\therefore$ Take $c = x_M$

Case 2 Both $x_m$ and $x_M \in a$ or $b$

Since $f(a)=f(b)$

$\rightarrow f(x_m)=f(x_M)$, i.e. Global Min = Global Max $\longrightarrow f(x) \equiv$ const $C$ $\Rightarrow f'(x)=0$ on $[a,b]$

equally everywhere on $[a,b]$

QED Take $c =$ any value on $[a,b]$

4.2/19 $f(x) = x^4+3x+1$ $[-2,-1]$

Existence of zero - $f$ is contin on $[-2,-1]$
- $f(-2)=11$, $f(-1)=-1$

By IVT, $\exists\, c \in (-2,-1)$ s.t. $f(c)=0$

Uniqueness of zero

$\rightarrow$ Suppose $\exists\, c_1, c_2 \in (-2,-1)$ s.t. $f(c_1)=0=f(c_2)$

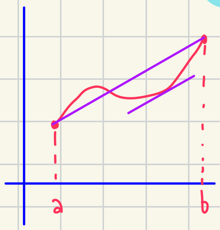
By Rolle's theorem with $a=c_1$, $b=c_2$, we have $\exists\, c \in (c_1,c_2)$ s.t. $f'(c)=0$

But $f'(x) = 4x^3+3 \overset{\text{set}}{=} 0$

$$x^3 = -\frac{3}{4}$$
$$x = \sqrt[3]{-\frac{3}{4}} = \boxed{c \overset{\text{supposed}}{\notin} (-2,-1)} \; \lightning \; / \rightarrow\leftarrow$$

$\therefore$ We don't have $c_1$ and $c_2$ QED

General case: **Mean Value Theorem**




Suppose $f$ is contin on $[a,b]$, differentiable in $(a,b)$, then $\exists\, c \in (a,b)$ s.t. $\frac{f(b)-f(a)}{b-a} = f'(c)$

Proof Eq. of secant line:
$$g(x) = y = f(a) + \left(\frac{f(b)-f(a)}{b-a}\right)(x-a)$$

Define $h(x) = f(x) - g(x) \rightarrow h$ continuous on $[a,b]$
$h$ differentiable on $(a,b)$

$$\left.\begin{array}{l} h(a) = f(a) - f(a) = 0 \\ h(b) = f(b) - (f(a) + (f(b)-f(a))) = 0 \end{array}\right\}$$ By Rolle's thm, $\exists\, c \in (a,b)$ s.t. $h'(c)=0$

$$f'(c) - g'(c) = 0$$
$$\frac{f(b)-f(a)}{b-a} = g'(c) = f'(c)$$

Corollary 1 Suppose $f' \equiv 0$ on interval $I$, then $f =$ const

Proof $\forall x_1, x_2 \in I$ s.t. $x_1 < x_2$ Want to show $f(x_1)=f(x_2)$

Apply MVT to $[x_1, x_2] \rightarrow \exists\, c \in (x_1,x_2)$ s.t. $f'(c) = \frac{f(x_2)-f(x_1)}{x_2-x_1} = 0$ $\therefore f(x_1)=f(x_2)$

Corollary 2 Suppose $f' \equiv g'$ on interval $I$, then $\exists$ const $C$ s.t. $f(x)-g(x) = C$ on $I$ i.e. $f(x) = g(x) + C$
(distance)

Proof Let $h = f - g$
$h' \equiv f' - g' \equiv 0$ on $I$
By corollary 1, $h \equiv$ const $C$ QED

4.2/31 b) $y' = x^2$

Special: $y = \frac{x^3}{3}$

Now if $y' = x^2 = y_1' \rightarrow y = y_1 + C$ $\leftarrow$ Generic case
$= \frac{x^3}{3} + C$

62. $\underset{-1}{a}$ ——— $\underset{1}{b}$

$$\frac{f(1)-f(-1)}{1-(-1)} = f'(c) \underset{<\frac{1}{2}}{\overset{>0}{}} \quad \exists c \in (-1,1)$$

$\to f(1) > f(-1)$

$\to f(1) - f(-1) < \frac{1}{2} \cdot 2 = 1$

$\therefore f(1) < f(-1) + 1$

## 4.3 Monotonic functions & first derivative test

Corollary 3 (of MVT) Suppose $f$ contin $[a,b]$ & diff $(a,b)$. If $f'>0$ on $(a,b)$ then $f$ is increasing on $[a,b]$, i.e. $a \le x_1 \le x_2 \le b$, $f(x_1) \le f(x_2)$

Proof Apply MVT on $[x_1, x_2] \to \frac{f(x_2)-f(x_1)}{x_2-x_1} = f'(c) > 0$ ; $c \in (x_1, x_2)$ QED

Q: Given $f(x)$ on $[a,b]$, how to find local extreme values of $f$ ?

General A: Candidates for location of local extreme values of $f$ are critical points and end points of $[a,b]$

More specific A: First derivative test for loc extrema: Suppose $c$ is critical point of $f$ and $f$ is continuous in neighborhood of $c$ and $f$ differentiable in truncated neighborhood

$( \underset{c}{|} )$

Sign of $f'$

1) $( f'>0 \underset{c}{|} f'<0 )$ $f(c)$ loc max

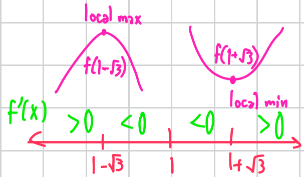
2) $( f'<0 \underset{c}{|} f'>0 )$ $f(c)$ loc min

3) $( f'<0 \underset{c}{|} f'<0 )$ $f(c)$ is increasing in nbhd

4) $( f'<0 \underset{c}{|} f'<0 )$ $f(c)$ is decreasing in nbhd

HW 3.9-4.3
+Quiz

35. $f(x) = \frac{x^2+2}{x-1}$




Step 1 Plot on x-axis the pts where $f$ is undefined $(x=1)$

Step 2 Find all c.p.s by solving $f'(x)=0$

$$\frac{(x-1)2x-(x^2+2)(1)}{(x-1)^2} = 0 \to x^2-2x-2=0$$

$$x = \frac{2 \pm \sqrt{4+8}}{2} = 1 \pm \sqrt{3}$$

Step 3 Pts obtained on Step 1,2 divide x-axis into several intervals, on each of which, $f'(x)$ should NOT change sign

Hence to find sign of $f'$ on each interval, just take a test point to see the sign of $f'$